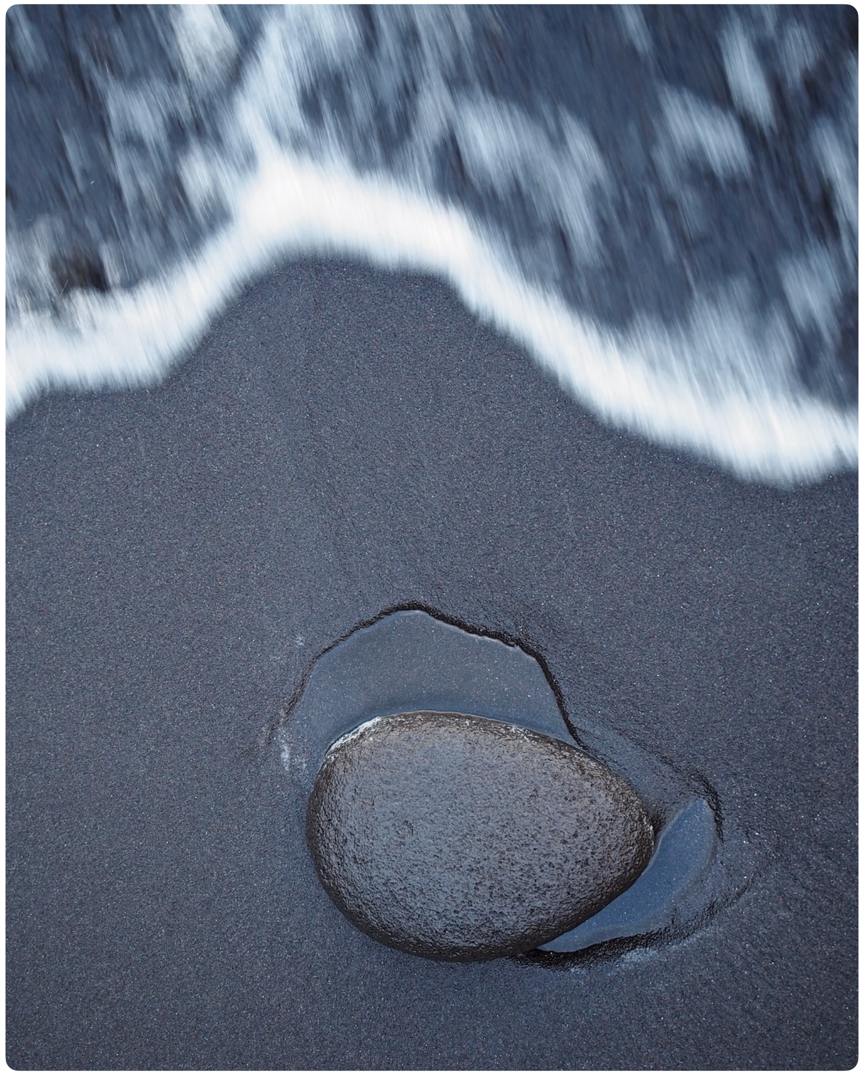
Exercises in Marine Ecological Genetics

05. Variant calling and SNPs

- VCF format
- Assessing SNP quality metrics
- Filtering SNP datasets
- Variant calling (optional)



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https://github.com/mhelmkampf/meg26

Today's objectives

By the end of this session, you will be able to:

- Understand the structure of **VCF files**, the main file format to store Single Nucleotide Polymorphism (SNP) data
- Calculate **summary statistics from SNP datasets**, including missingness, sequencing depth, and allele frequency distributions
- **Filter and manipulate SNP datasets** using standard tools (VCFtools), and explain the rationale behind common filtering decisions
- Trace the origin of a VCF file by **calling variants** from raw sequencing reads (optional)

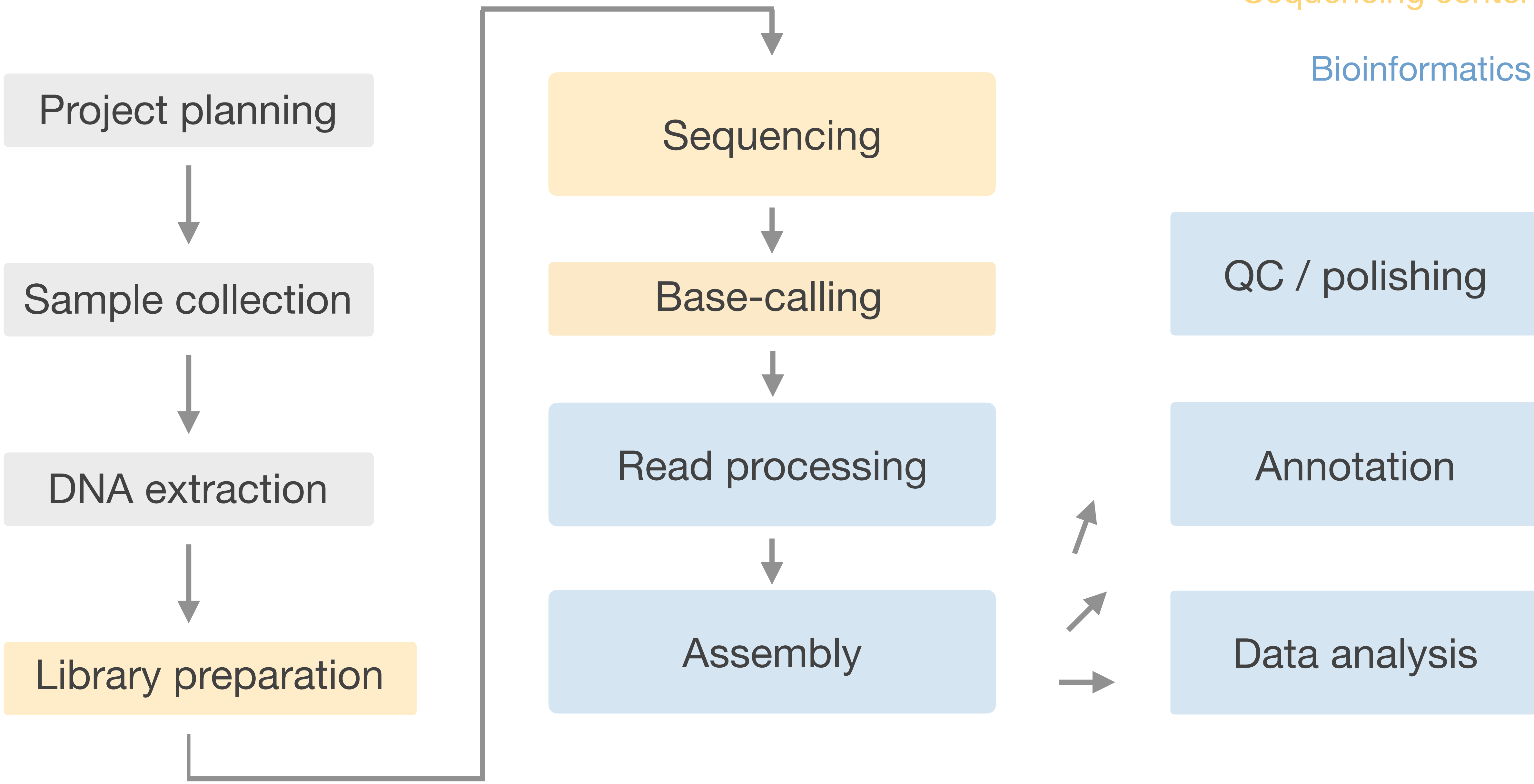
De novo genome sequencing workflow

Legend

Preparation

Sequencing center

Bioinformatics



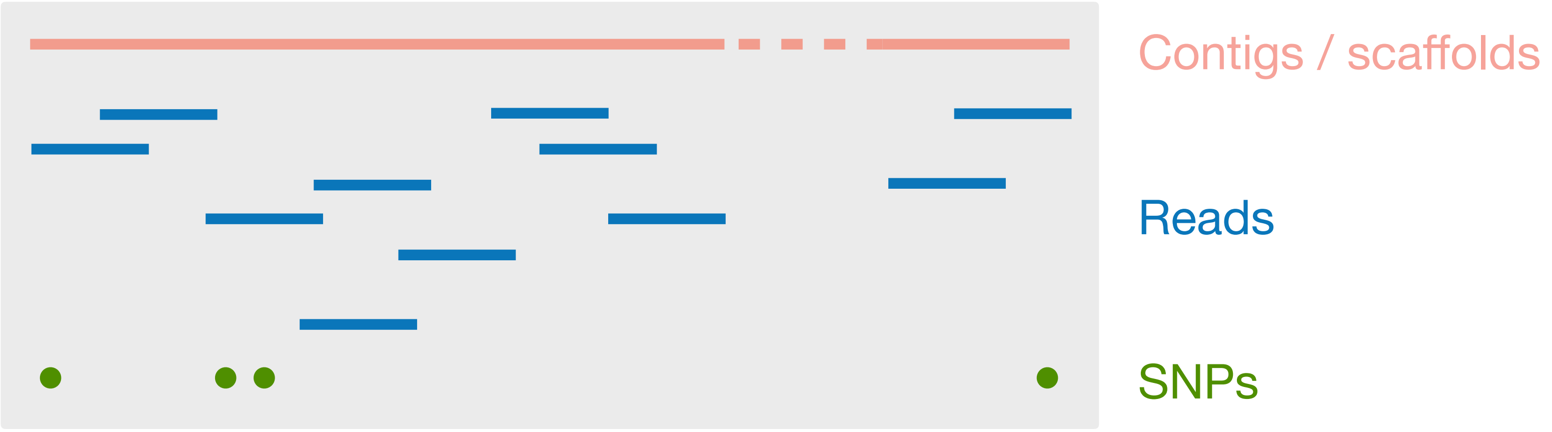
Genome sequencing strategies

De novo

→ High-quality reference genome

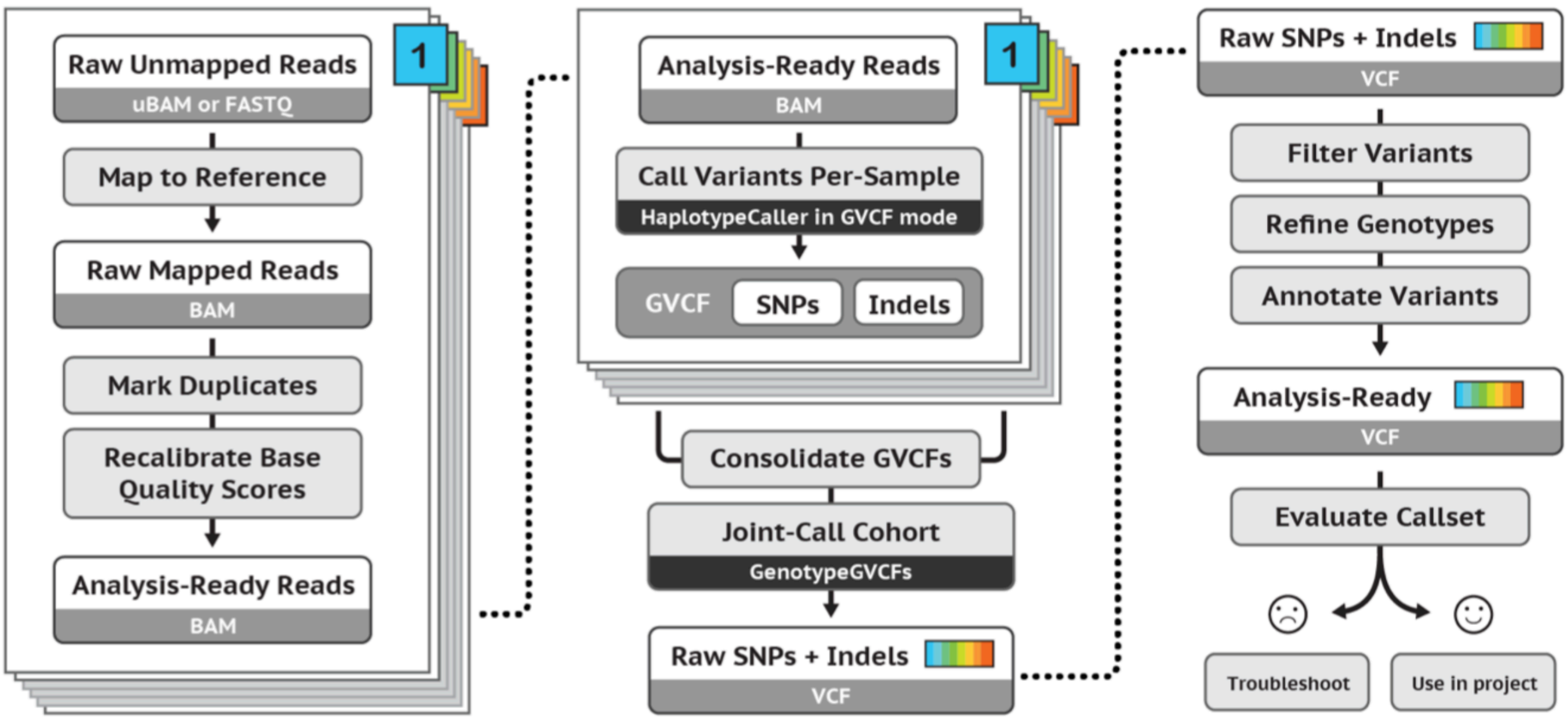
Re-sequencing

→ SNP dataset for population genomics



More affordable alternative: reduced representation sequencing, e.g. RADseq

Variant calling workflow (GATK software)



Read processing / mapping

Variant calling / genotyping

Filtering / QC

gatk.broadinstitute.org

Read mapping

- Aligns short sequencing reads to a reference genome to determine their most likely genomic origin
- Repetitive regions, duplicated genes or sequencing errors can cause poor mapping
- Alignments are typically stored in SAM/BAM files that include mapping quality scores and form the basis for variant calling and genotyping

```
G A T G T T C G A A
G A T C G A A
G A C C T C G T
        ↓
G A T G T T C G A A
G A T C - - - G A A
G A C C - T C G - T
```

Alignment

arranges nucleotide or amino acid sequences so that the number of mismatches and gaps are minimized

Sequence Alignment Map (SAM) format

```
samtools view -F 4 indbel-mtg_unsorted.sam | head -n 1  # print 1st mapped read
```

```
E00489:149:H3H77CCXY:7:1101:27783:2364            99          LG_M
11013       60          150M        =           11247       360
GATAAAAAGACTAATTGTTTCGATGACAATCAGGACAGGAATTAGAGGGCCGGGGGTTCCTTCTGGAAGAAGATGGCCTA
ACGCGTGAGTTGGCTGATTACGCATTCCAATTAGGACGGTTGCTAGTCATAGGGGGGTTGCAATTCCAAG
AAAFFJJJJJJJJJAJJJJ-7FJAFJFJJJJAJJJJJJJJJJJJJJJFJJJJJJJJ7A7FJFJ<FJJJJJJJJJJ7FFFJ
JJJAJ7AFF-7FFJJFJJJJ77J7-A-FJF<FJ7F7AJJ--FA-F---FF<AAAJJA))AFFA--<F7J7
NM:i:2      MD:Z:60C82G6 MC:Z:126M   AS:i:140    XS:i:0
```

Key steps in genotyping

1. **Genotype likelihood calculation**
    - Base calls (A, C, G, T)
    - Base quality scores
    - Mapping quality
    - Read depth
    - ▶ Probabilities for each genotype (e.g. AA, AC, CC)

2. **Genotype inference**
    - Genotype likelihoods
    - Other probabilities (e.g. mutation rate, population-level information)
    - Statistical inference models
    - ▶ VCF file with genotype calls and confidence metrics

The Variant Call Format (VCF)

Header (meta information)
##
column names

Columns are individuals

Rows are sites (variants, SNPs)

Site information
(fixed 9 columns)

Genotypes

Efficiently stores

genetic variation data

Optional data e.g:

- allele frequencies
- no. samples with data
- read depth
- genotype likelihood

Variant quality and read depth

Phred quality score: $Q = -10 \log_{10} P$

Quality score	Prob. site is NOT variant	Accuracy
10	1 in 10	90 %
20	1 in 100	99 %
30	1 in 1000	99,9 %
40	1 in 10000	99,99 %
…	…	…

Read depth:

```
Genome: CGTAATGGCATATCGCCTAGATTCGAAACG
Read 1:   TAATGGCATATCGCCTAGAT
Read 2:         CATATCGCCTAGATTCGAAA
Read 3:           TATCGCCTAGATTCGAAACG
Depth:  001111112233333333333322222211
```

Variant calling and SNPs

Summary

```
zcat input.vcf.gz | less        # inspect compressed VCF file

vcftools \                      # filter variants by …
  --minQ 40 \                   # quality
  --minDP 10 \                  # depth
  --max-missing 0.8 \           # missingness
  --max-alleles 2 \             # num. of alleles
  --mac 2 \                     # rarity
  --recode \
  --stdout | bgzip > filtered.vcf.gz

bcftools stats input.vcf.gz     # VCF summary statistics (num. of sites, indiv.)
```

Key takeaways

Summary

- **VCF files** are a widely used and efficient format for storing and handling genetic variant data, especially SNPs
- SNP datasets **require careful quality control** before analysis, including filtering for sequencing depth, missing data, and allele frequency to ensure reliable genotypes
- Variant calling from raw reads involves many steps that are performed in **complex pipelines** that may introduce technical biases if not properly understood
- High-quality SNP datasets form the foundation for many downstream analyses in **population genomics** to study population differentiation, adaptation, demographic history, gene flow etc.